

Introduction to numerical methods - 4 -

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Gauss - Elimination

$$\underline{A} \underline{x} = \underline{b}$$

- Swapping rows $\rightarrow$ pivoting
- Multiply row by a non-zero number
- Add a row to another

Used to avoid zeros in the diagonal

pros:

- general
- exact results (not iterative)
- simple algorithm

cons:

- computational cost $O(n^3)$
- instability (rounding, small and large numbers)
- inefficient for sparse matrices

P Pivoting matrix

Start from $\underline{P} \equiv \underline{I}$ when row i and j are exchanged change rows in $\underline{P}$ too

e.g. row 1 & row 2

$$\underline{P} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \hat{P}$$

LU decomposition

$$\underline{P} \underline{A} = \underline{L} \underline{U} = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} \cdot \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}$$

$\underline{L}$ and $\underline{U}$ can be obtained from other algorithms, e.g. Doolittle, Crout's, or from Gauss elimination.

Iterative methods: Gauss-Seidel

Let us have $\underline{A}$ with non-zero diagonal elements $a_{ii} \neq 0$

We express x_i from the i th eq. and use it as a new value in the iteration: (iteration $t+1$)

$$x_i^{(t+1)} = \frac{-1}{a_{ii}} (a_{i2} x_2^{(t)} + \dots + a_{in} x_n^{(t)} - b_i)$$

$$\vdots$$

$$x_n^{(t+1)} = \frac{-1}{a_{nn}} (a_{n1} x_1^{(t)} + \dots + a_{n,n-1} x_{n-1}^{(t)} - b_n)$$

This is the Jordan method. Iterative methods work better (in general) if they are complicate.

$$x_1^{(t+1)} = -\frac{1}{a_{11}} (a_{12}x_2^{(t)} + \dots + a_{1n}x_n^{(t)} - b_1)$$

$$x_2^{(t+1)} = -\frac{1}{a_{22}} (a_{21}x_1^{(t+1)} + a_{23}x_3^{(t)} + \dots + a_{2n}x_n^{(t)} - b_2)$$

$\vdots$

$$x_n^{(t+1)} = -\frac{1}{a_{nn}} (a_{n1}x_1^{(t+1)} + \dots + a_{nn-1}x_{n-1}^{(t+1)} - b_n)$$

or

$$x_i^{(t+1)} = -\frac{1}{a_{ii}} \left(\sum_{j=1}^{i-1} a_{ij}x_j^{(t+1)} + \sum_{j=i+1}^n a_{ij}x_j^{(t)} - b_i \right)$$

Only works if A is either

- diagonally dominant $|a_{ii}| \geq \sum_{j \neq i} |a_{ij}|$
- positive definite
- symmetric

Matrix inverse with Gauss-Jordan

$$\begin{array}{l} \begin{pmatrix} \underline{A} & | & \underline{I} \end{pmatrix} \xrightarrow{GE} \begin{pmatrix} \underline{I} & | & \underline{A}^{-1} \end{pmatrix} \\ \uparrow \\ \begin{pmatrix} a_{11} & a_{12} & a_{13} & | & 1 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & | & 0 & 1 & 0 \\ a_{31} & a_{32} & a_{33} & | & 0 & 0 & 1 \end{pmatrix} \end{array} \quad \left| \begin{array}{l} \text{Proof: GE is a lin. op.} \\ \underline{E}: \underline{E} \cdot \underline{A} = \underline{I} \rightarrow \underline{E} = \underline{A}^{-1} \\ \downarrow \\ \underline{E}(\underline{A} | \underline{I}) \rightarrow (\underline{I} | \underline{E}\underline{I}) = \\ = (\underline{I} | \underline{E}) \quad \text{Q.E.D.} \end{array} \right.$$

Eigen values, vectors

$$\underline{A} \underline{v} = \lambda \underline{v}$$

Standard method

- eigen values: roots of the characteristic polynomial

- eigen vectors: solution of a singular set of linear equations

e.g. $\underline{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

$$\det(\underline{A} - \lambda \underline{I}) = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = 3 - 4\lambda + \lambda^2 = 0$$

$$\lambda_1 = 1 \quad \lambda_2 = 3$$

$$\underline{v}_1: (\underline{A} - \lambda_1 \underline{I}) \underline{v}_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_{11} \\ v_{12} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad v_{11} + v_{12} = 0$$

$$\underline{v}_1 = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\underline{v}_2: (\underline{A} - \lambda_2 \underline{I}) \underline{v}_2 = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_{21} \\ v_{22} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad v_{21} - v_{22} = 0$$

$$\underline{v}_2 = c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Power Method: finding the largest eigenvalue

A $n \times n$ $\lambda_1, \lambda_2, \dots, \lambda_n$ eigenvalues such that $|\lambda_1| > |\lambda_2| > \dots > |\lambda_n|$ with corresponding eigenvectors $\underline{v}_i$.

Start from an arbitrary vector

$$\underline{x}_0 = c_1 \underline{v}_1 + c_2 \underline{v}_2 + \dots + c_n \underline{v}_n \quad \text{with } c_1 \neq 0$$

$$\underline{A} \underline{x}_0 = \sum_i c_i \underline{A} \underline{v}_i = \sum_i c_i \lambda_i \underline{v}_i =$$

$$= c_1 \lambda_1 \underline{v}_1 + c_2 \frac{\lambda_2}{\lambda_1} \underline{v}_2 + \dots + c_n \frac{\lambda_n}{\lambda_1} \underline{v}_n$$

$$\underline{A}^n \underline{x}_0 = \lambda_1^n \left(c_1 \underline{v}_1 + c_2 \frac{\lambda_2^n}{\lambda_1^n} \underline{v}_2 + \dots + c_n \frac{\lambda_n^n}{\lambda_1^n} \underline{v}_n \right)$$

if n is sufficiently large $\frac{\lambda_i^n}{\lambda_1^n} \ll 1$ $i > 1$

$$\Rightarrow \underline{A}^n \underline{x}_0 \approx \lambda_1^n c_1 \underline{v}_1$$

Method: start from $\underline{b}_0$ $\underline{b}_0$ must

$$\underline{b}^{(k+1)} = \frac{\underline{A} \underline{b}^{(k)}}{\|\underline{A} \underline{b}^{(k)}\|}$$

have nonzero component in $\underline{v}_1$ dir.

Eigenvalue shift: subtract $\underline{\ell \cdot \underline{I}}$ from $\underline{A}$

$$\underline{A'} = \underline{A} - \underline{\ell \cdot \underline{I}} \Rightarrow \lambda'_i = \lambda_i - \ell$$

eigenvalues shift by ℓ

Usage:

- precise eigenvalue from 'close enough' guess
 - + subtract the close enough value
 - + invert the matrix
 - + do the iteration
- We will get $(\lambda_i - \ell)^{-1}$ as the largest eigenvalue